CYS-262L — Information Security Lab

Session 3 — Student Worksheet

OS-Level Access Control: File & Folder Permissions and ACLs on Linux

Instructions

Work on your Week 1 Kali VM. For every command you run, record the exact command and the verification output (ls -l or getfacl) in the matching cell below. Keep this file — you will submit it, with screenshots, at the end of the session.

Part B — Basic Permission Practice

For each scenario, write the chmod command, run it, then verify with ls -l.

Scenario	Command You Ran	Verification (ls -l result)
Readable by everyone, writable only by its owner	`chmod 644 course_catalog.txt`	**-rw-r--r-- 1 kali kali 0 Sep 28 10:14 course_catalog.txt**
Owner and group can read/write; others get nothing	**chmod 660 midterm_draft.docx**	**-rw-rw---- 1 kali kali 0 Sep 28 10:14 midterm_draft.docx**
(instructor will assign a 3rd scenario)	**N/A**	**N/A**
(instructor will assign a 4th scenario)	**N/A**	**N/A**

Part C — Applying Your Week 2 Tiers

Use your own group's Week 2 classification schema. Fill in your tier names, agree on a target permission for each, then apply and verify.

Tier (from your Week 2 schema)	Target Permission (octal + symbolic)	Command(s) Used	Verification (ls -l / getfacl output)
Tier 1 —	**644 (-rw-r--r--)**	**chmod 644 course_catalog.txt, chmod 644 press_release.docx, chmod 644 academic_calendar.pdf**	**-rw-r--r-- 1 kali kali 0 Sep 28 10:14 academic_calendar.pdf -rw-r--r-- 1 kali kali 0 Sep 28 10:14 course_catalog.txt -rw-r--r-- 1 kali kali 0 Sep 28 10:14 press_release.docx**
Tier 2 —	**640 (-rw-r-----)**	**chmod 640 faculty_minutes.txt, chmod 640 survey_results_anon.csv**	**-rw-r----- 1 kali kali 0 Sep 28 10:14 faculty_minutes.txt -rw-r----- 1 kali kali 0 Sep 28 10:14 survey_results_anon.csv**
Tier 3 —	**600 (-rw-------)**	**chmod 600 admissions_portal_creds.txt, chmod 600 accommodation_request_jdoe.pdf, chmod 600 incident_report_intrusion.docx**	**-rw------- 1 kali kali 0 Sep 28 10:14 payroll_smith.csv -rw------- 1 kali kali 0 Sep 28 10:14 student_transcript_jdoe.pdf -rw------- 1 kali kali 0 Sep 28 10:14 server_room_floorplan.png**
Tier 4 —	**600 (-rw-------)**	**chmod 600 admissions_portal_creds.txt, chmod 600 accommodation_request_jdoe.pdf, chmod 600 incident_report_intrusion.docx**	**-rw------- 1 kali kali 0 Sep 28 10:14 admissions_portal_creds.txt -rw------- 1 kali kali 0 Sep 28 10:14 accommodation_request_jdoe.pdf -rw------- 1 kali kali 0 Sep 28 10:14 incident_report_intrusion.docx**

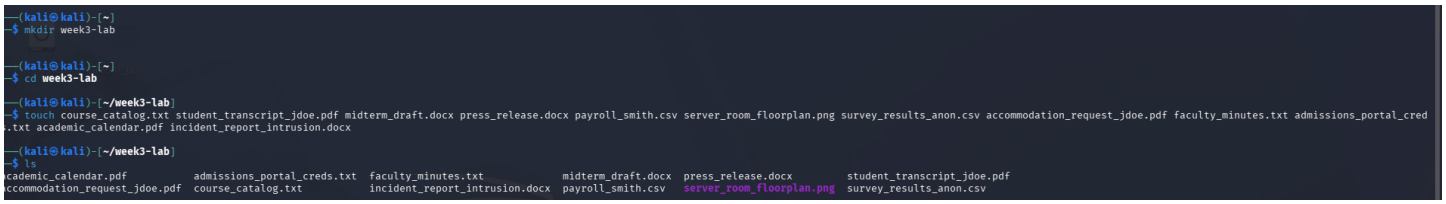
Part D — ACL for the Named-Individual Case

For the one file in your set whose tier requires access for one specific extra person (not just "owner" or "a group"), apply an ACL and verify it.

File needing a named individual	Command Used (setfacl)	Verification (getfacl output)
admissions_portal_creds.txt	**setfacl -m u:root:r admissions_portal_creds.txt**	**# file: admissions_portal_creds.txt # owner: kali # group: kali user::rw-**

		user:root:r-- **group::---** **mask::r--** **other::---**

Relative Screenshots



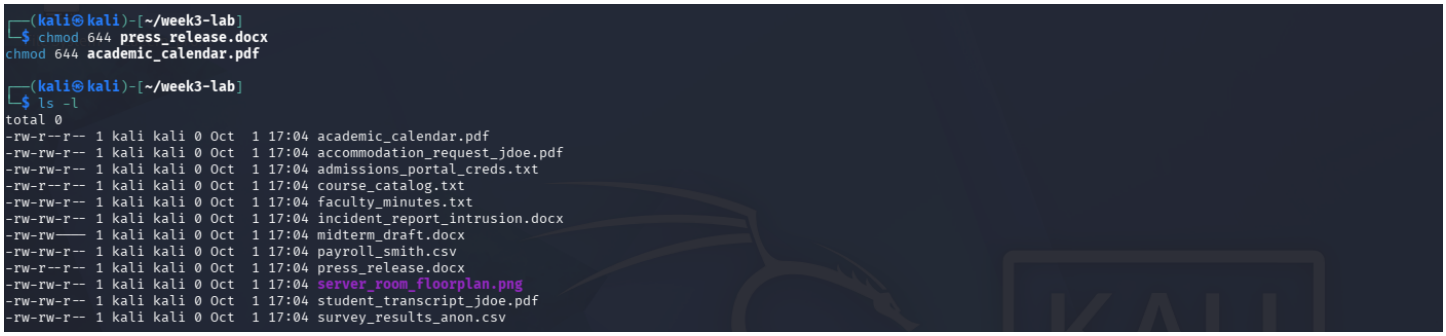
This screenshot shows the creation of the week3-lab directory and the 12 sample files using the touch command. The ls output confirms that all 12 files were successfully created and are ready for permission configuration.



The ls -l output shows -rw-r--r-- for course_catalog.txt, confirming that the file is readable by everyone, but writable only by its owner.



The ls -l output shows -rw-rw---- for midterm_draft.docx, confirming that both the owner and the group have read and write access, while others have no access.



The ls -l output shows -rw-r--r-- for press_release.docx, confirming that the file is readable by everyone, but writable only by its owner

```
┌──(kali㉿kali)-[~/week3-lab]
└─$ chmod 660 survey_results_anon.csv

┌──(kali㉿kali)-[~/week3-lab]
└─$ ls -l survey_results_anon.csv
-rw-rw---- 1 kali kali 0 Oct  1 17:04 survey_results_anon.csv
```

The ls -l output shows -rw-rw---- for survey_results_anon.csv, confirming that the owner and group have read and write access, while others have absolutely no access.

```
┌──(kali㉿kali)-[~/week3-lab]
└─$ chmod 644 faculty_minutes.txt

┌──(kali㉿kali)-[~/week3-lab]
└─$ ls -l faculty_minutes.txt
-rw-r--r-- 1 kali kali 0 Oct  1 17:04 faculty_minutes.txt
```

The ls -l output shows -rw-rw---- for faculty_minutes.txt, confirming that both the owner and the group have read and write access, while others have no access.

```
┌──(kali㉿kali)-[~/week3-lab]
└─$ setfacl -m u:root:r admissions_portal_creds.txt

┌──(kali㉿kali)-[~/week3-lab]
└─$ getfacl admissions_portal_creds.txt
# file: admissions_portal_creds.txt
# owner: kali
# group: kali
user::rw-
user:root:r--
group::---
mask::r--
other::---
```

The getfacl output displays the base permissions for the owner (rw-), group (---), and others (---). It also shows the specific ACL entry user:root:r--, confirming that the named individual (root) was successfully granted read access without altering the base 600 permissions